

Hitesh Gupta

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SUMMARY

Undergraduate student in Artificial Intelligence and Machine Learning with hands-on experience across 3 end-to-end projects in computer vision, deep learning, and predictive modeling. Proficient in Python, NumPy, pandas, Scikit-learn, and OpenCV, with applied experience in object detection, image processing, model deployment, REST APIs, and generative adversarial networks (GANs).

EDUCATION

Vellore Institute of Technology September 2023 – May 2027
B.Tech in Computer Science and Engineering (AI & ML) CGPA: 8.68

DS Science Academy July 2021 – April 2022
Class XII Percentage: 89.80%

EXPERIENCE

Edunet Foundation (in association with Microsoft and AICTE) June 2025 – July 2025
AI and Machine Learning Intern Remote

- Completed practical labs on Microsoft Learn and Edunet LMS covering supervised and unsupervised learning, classification, regression, and unlabeled data handling.
- Worked with Azure Custom Vision for image classification and applied Generative AI concepts through guided demonstrations on Microsoft Azure.

PROJECTS

Smart Retail Checkout System | Python, YOLOv8, FastAPI, PyTorch **April 2026**

- Built an AI-based retail billing system by training a YOLOv8s model on the IIT Patna Grocery Dataset (5,983 images, 1,816 product classes) for 100 epochs, achieving mAP@50 of 0.285 across Indian retail brands including Amul, Maggi, and Haldiram's.
- Developed a FastAPI backend for real-time inference and a React.js frontend supporting multi-image upload, confidence-based filtering, automatic itemized bill generation with GST calculation, and print export.
- Addressed class imbalance (avg. 3.3 samples per class), mixed annotation formats, and version compatibility issues across PyTorch and Ultralytics training environments.

Lane Detection and Driver Assistance System | Python, OpenCV, Ultralytics YOLO11 **January 2026**

- Built a computer vision pipeline for lane segmentation and steering angle estimation, training a YOLO11n-seg model on the Indian Driving Dataset (IDD) with 6,993 annotated road images.
- Engineered a mask-to-YOLO annotation pipeline using OpenCV contour detection to convert segmentation masks into polygon labels, enabling an 80/20 train-validation split for supervised training.
- Designed a geometric steering angle algorithm ($\pm 30^\circ$ range) from lane-centroid offset and implemented inference scripts for dataset and external images with visual overlays for interpretability.

Human Sustainability Score Prediction Model | Python, Scikit-learn, pandas, NumPy **May 2025**

- Designed a modular pipeline with four components – Electricity, Transport, Recycling, and Tech – to predict individual sustainability scores by converting lifestyle inputs into standardized CO2 equivalents using logistic scaling functions.
- Trained and evaluated six machine learning models, including Linear Regression, Random Forest, and Gradient Boosting; Linear Regression achieved the best performance ($R^2 = 0.68$, $MAE = 0.37$).
- Visualized model performance through MSE comparisons, residual error distributions, and score distribution graphs to support data-driven environmental policymaking.

TECHNICAL SKILLS

Languages: Python, Java, SQL

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab

Technologies/Frameworks: NumPy, pandas, Scikit-learn, Matplotlib, OpenCV, PyTorch, Ultralytics YOLO, FastAPI

ACHIEVEMENTS

Certifications: Applied Machine Learning in Python (Coursera); Java Programming (HackerRank); Data Analytics Job Simulation (Deloitte Virtual Experience Program)

Competitive Programming: Solved 300+ algorithmic problems across LeetCode, CodeForces, and GeeksforGeeks.